DigitalTwin.ai — Project Overview

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🎯 What Is It?

DigitalTwin.ai is a **Spatio-Temporal Graph Intelligence** system for vehicle assembly lines. It models an entire 40-station production line as a **dynamic directed graph** — stations are nodes, material/information flows are edges — powered by a **Spatio-Temporal Graph Neural Network (ST-GNN)**.

🏗️ Architecture (3-Layer)

```
graph LR
    A["DATA LAYER<br/>OPC-UA / MQTT<br/>Edge Gateways<br/>Sensor Masking"] --> B["INTELLIGENCE
    B --> C["APPLICATION LAYER<br/>Multi-Stakeholder Dashboard<br/>REST API<br/>Real-Time Alerts
```

📊 Node Features (12 per Station)

Each of the 40 stations (nodes) has **12 input features** that represent its real-time operational state:

#	Feature	Unit	Description	Always Observable?
0	`queue_length`	count	Number of units waiting at the station	✅ Yes (visual count)
1	`cycle_time`	seconds	Actual processing time per unit (×takt)	✅ Yes (PLC/manual)
2	`defect_rate`	%	Rate of quality defects at the station	✅ Yes (inspection records)
3	`temperature`	°C	Station/process temperature	⚠️ FULL & PARTIAL only
4	`vibration`	mm/s	Vibration measurement from equipment	🔴 FULL only
5	`torque_deviation`	%	Deviation from target torque	🔴 FULL only

#	Feature	Unit	Description	Always Observable?
6	`utilization`	%	Equipment uptime percentage	✅ Yes
7	`throughput`	fraction	Output as fraction of target	✅ Yes
8	`humidity`	%	Ambient humidity (critical in paint zone)	⚠️ FULL & PARTIAL only
9	`power_draw`	kW	Electrical power consumption	🔴 FULL only
10	`hrs_since_maint`	hours	Hours since last maintenance window	✅ Yes (logs)
11	`operator_skill`	index 0–1	Operator experience/training level	✅ Yes (HR records)

Instrumentation Levels & Feature Observability

Level	% of Stations	Features Available	Missing (must be inferred by GNN)
FULL	60% (24 stations)	All 12	None
PARTIAL	25% (10 stations)	9 of 12	vibration, torque_deviation, power_draw
MANUAL	15% (6 stations)	8 of 12	temperature, vibration, torque_deviation, humidity

🏭 40-Station Graph: 3 Production Zones

Zone A — Body Construction (13 stations, IDs 0–12)

ID	Station	Short	Instrumentation	Vintage	Takt (s)
0	Blanking Press	BLANK	FULL	Modern	45
1	Stamping Press A	STMP-A	FULL	Modern	52
2	Stamping Press B	STMP-B	FULL	**Legacy**	55
3	Laser Cutting	LASER	FULL	Modern	38
4	Hemming Station	HEM	**PARTIAL**	**Legacy**	60
5	Spot Weld Cell 1	WELD-1	FULL	Modern	48
6	Spot Weld Cell 2	WELD-2	FULL	Modern	48
7	Spot Weld Cell 3	WELD-3	**PARTIAL**	**Legacy**	55
8	Adhesive Bonding	BOND	**PARTIAL**	**Legacy**	42
9	Geometry Check	GEO	FULL	Modern	35
10	BiW Audit	BIW-AUD	**MANUAL**	**Legacy**	90

ID	Station	Short	Instrumentation	Vintage	Takt (s)
11	Sealer Application	SEAL	**PARTIAL**	**Legacy**	50
12	Underbody Coating	UCOAT	FULL	Modern	40

Zone B — Paint Shop (11 stations, IDs 13–23)

ID	Station	Short	Instrumentation	Vintage	Takt (s)
13	Pretreatment Bath	PRETR	FULL	Modern	120
14	E-Coat Dip Tank	ECOAT	FULL	Modern	180
15	E-Coat Cure Oven	EC-CURE	FULL	Modern	200
16	Seam Seal Booth	SEAM	**PARTIAL**	**Legacy**	65
17	Primer Spray Booth	PRIMER	FULL	Modern	55
18	Primer Cure Oven	PR-CURE	**PARTIAL**	**Legacy**	180
19	Color Spray Booth	COLOR	FULL	Modern	60
20	Color Cure Oven	CL-CURE	FULL	Modern	180
21	Clear Coat Booth	CLEAR	FULL	Modern	55
22	Clear Cure Oven	CC-CURE	**PARTIAL**	**Legacy**	180
23	Paint Inspection	P-INSP	**MANUAL**	**Legacy**	120

Zone C — Final Assembly (16 stations, IDs 24–39)

ID	Station	Short	Instrumentation	Vintage	Takt (s)
24	Wiring Harness	WIRE	**MANUAL**	**Legacy**	95
25	Dashboard Install	DASH	**PARTIAL**	**Legacy**	110
26	Headliner Install	HEAD	**MANUAL**	**Legacy**	75
27	Door Trim Assembly	DTRIM	**MANUAL**	**Legacy**	85
28	Glass Installation	GLASS	FULL	Modern	70
29	Seat Installation	SEAT	FULL	Modern	65
30	Chassis Marriage	MARRY	FULL	Modern	80
31	Engine/Motor Drop	ENGN	FULL	Modern	90
32	Suspension Torque	SUSP	FULL	Modern	70
33	Brake Line Connect	BRAKE	**PARTIAL**	**Legacy**	55
34	Fluid Fill Station	FLUID	FULL	Modern	50
35	Wheel Mounting	WHEEL	FULL	Modern	45

ID	Station	Short	Instrumentation	Vintage	Takt (s)
36	Battery/Fuel System	BATT	FULL	Modern	60
37	Final Torque Audit	TORQ-A	FULL	Modern	55
38	ADAS Calibration	ADAS	FULL	Modern	120
39	Roll Test & Dispatch	ROLL	**MANUAL**	**Legacy**	180

🔗 Graph Topology (48 Edges)

```
graph TD
    subgraph EdgeTypes["Edge Types"]
        FW["Forward Flow (36 edges)<br/>Material moves downstream"]
        FB["QC Feedback (10 edges)<br/>Inspection signals upstream"]
        RW["Rework Loops (2 edges)<br/>Defectives return upstream"]
    end
```

Edge Type	Count	Purpose	Example
Forward flow	36	Material moves station to station	Blanking → Stamping A
QC Feedback	10	Inspection results flow upstream	Geometry Check → Weld Cells
Rework	2	Failed units re-enter upstream	BiW Audit fail → Spot Weld 1

🧠 What We Used for What

Model Architecture — model.py

Component	Technology	Purpose
GATv2Conv (4 attention heads)	PyTorch Geometric	**Spatial reasoning** — learns which neighboring stations influence each other via graph attention. Captures station-to-station dependencies across the assembly line
GRU Cell (per node)	PyTorch	**Temporal reasoning** — captures time-series dynamics at each station (shift ramp-up, degradation, break patterns)
3 × Spatio-Temporal Blocks	Custom	Stacked GATv2+GRU blocks for deep spatial-temporal feature extraction
Input Projection (Linear → ELU → LayerNorm)	PyTorch	Maps 12 raw features → 128-dim hidden space
LayerNorm + Residual Connections	PyTorch	Training stability and gradient flow

5 Prediction Heads — What Each Does

Head	Type	Output	What It Tells You
Queue Forecast	Regression	`[40, 1]`	Predicted queue build-up at each station — warns of bottlenecks 10–15 min ahead
Defect Risk	3-class classification	`[40, 3]` (LOW/MEDIUM/HIGH)	Risk of defect propagation at each station
Virtual Sensor	Feature reconstruction	`[40, 12]`	Infers missing sensor readings for PARTIAL/MANUAL stations via neighbor aggregation
Anomaly Score	Binary probability	`[40, 1]` (0–1)	Real-time anomaly detection per station
Root-Cause Attribution	Attention weights	Per-edge weights	Identifies *which upstream station* is the root cause of downstream issues

Confidence Head

Component	Purpose
Confidence Score	Outputs per-node confidence (0–1). FULL-instrumented stations get ~0.90 base, PARTIAL ~0.65, MANUAL ~0.35. Flags when virtual sensor predictions are unreliable.

Simulator — simulator.py

Component	Purpose
Normal Baseline Generation	Generates realistic telemetry per station within defined operating ranges
Temporal Dynamics	Simulates shift ramp-up (first 10%), break period (~40–45%), end-of-shift fatigue (last 15%)
Anomaly Injection	5 built-in multi-causal scenarios injected at configurable time steps
Cascade Propagation	BFS up to 3 hops downstream — anomalous stations affect neighbors with attenuated signals
Sensor Masking	Zeros-out unobservable features for PARTIAL/MANUAL stations, forcing the GNN to infer them

Web Dashboard — app.py + dashboard.js

Component	Technology	Purpose
Backend	Flask (Python)	REST API serving predictions, timeline, what-if results, validation metrics
Frontend	Vanilla JS + CSS	Interactive multi-stakeholder dashboard with 3 views

Component	Technology	Purpose
What-If Engine	`model.simulate_intervention()`	Clones node features → applies hypothetical changes → re-runs GNN → shows predicted downstream impact

🚨 5 Built-In Anomaly Scenarios

#	Scenario	Category	Station(s)	Onset Step	Severity	What Happens
1	Weld Gun Electrode Wear	Equipment	WELD-3 (ID 7)	Step 5	0.7	Vibration ↑, cycle time ↑, defects ↑ — detected downstream at Geometry Check
2	Paint Booth Humidity	Environmental	COLOR, CLEAR (19, 21)	Step 8	0.6	Temperature ↑, humidity ↑ → orange peel & fish-eye defects
3	Stamping Die Wear	Equipment	STMP-B (ID 2)	Step 3	0.5	Dimensional drift propagates through welding stations
4	Shift Change Operator Drop	Operator	WIRE, DASH, HEAD, DTRIM (24–27)	Step 15	0.4	Skill ↓, cycle time ↑, defects ↑, queues ↑
5	Upstream Part Quality	Supply Chain	BOND (ID 8)	Step 10	0.6	Defect rate ↑, queue ↑, torque deviation ↑

📊 Three Stakeholder Views

View	Target User	Key Panels
Floor Supervisor	Real-time operator	Live assembly graph (color-coded risk), active alerts, station status matrix, virtual sensing panel
Plant Manager	Shift planner	Shift KPIs, risk heatmap (station × time), bottleneck frequency chart, what-if simulator, anomaly tracker
Leadership	Executives	ROI projections, model validation metrics (precision/recall/F1), multi-plant rollout plan, implementation roadmap

🎯 Outcomes & Business Impact

Metric	Expected Improvement
Unplanned Downtime	30–45% reduction
Rework Costs	25–35% reduction
Bottleneck Detection	From reactive → **10–15 min advance warning**
Sensor Hardware Costs	40–60% reduction via virtual sensing
Annual Savings (per plant)	**$2.8M–$4.5M**
Pilot Payback Period	~6–9 months

Technical Validation Metrics (Live)

The dashboard computes these against simulator ground truth in real-time: - **Precision** — How many flagged anomalies are real - **Recall** — How many real anomalies are caught - **F1 Score** — Harmonic mean of precision and recall - **False Alarm Rate** — Fraction of false positives

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🛠️ Tech Stack Summary

Layer	Technology	Why
Deep Learning	PyTorch 2.0+	Industry-standard DL framework, GPU support
Graph Neural Networks	PyTorch Geometric (GATv2Conv)	State-of-the-art graph attention for spatial dependencies
Graph Construction	NetworkX	Building & analyzing directed graph topology
Data Simulation	NumPy	Fast numerical computation for synthetic telemetry
Web Backend	Flask	Lightweight Python web server for REST API
Web Frontend	Vanilla JS + CSS	Interactive dashboard with no framework overhead
Visualization	Matplotlib	Static demo visualizations (4 output PNGs)

📁 File Map

File	Role	Lines
`model.py`	ST-GATv2 + GRU neural network with 5 prediction heads	279

File	Role	Lines
`assembly_line.py`	40-station definition, graph topology, sensor masks	411
`simulator.py`	Synthetic telemetry generation + anomaly injection	350
`app.py`	Flask backend — inference engine + REST API	381
`demo.py`	Standalone terminal demo with 4 visualizations	893
`templates/index.html`	Dashboard SPA template	—
`static/dashboard.js`	Frontend logic (graph, KPIs, charts)	—
`static/style.css`	Dark-themed dashboard styling	—